Paper Replication Report #067: Exoplanet Phase Curves & Day-Night Heat Redistribution

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Cowan & Agol (2011) and Showman et al. (2009) modeling thermal phase curves of short-period tidally-locked giant exoplanets, day-night temperature contrast $T_{\text{day}}/T_{\text{night}}$, atmospheric recirculation efficiency $\mathcal{E}_{\text{recirc}}$, and phase offsets $\Delta\phi$ caused by eastward equatorial superrotating jets. Using our C++ solver engine, we evaluate phase flux variations $F(\phi)$ across orbital phases $\phi \in [0^\circ, 360^\circ]$. Numerical phase curves match Spitzer and Hubble phase space observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Tidally locked Hot Jupiters undergo intense unilateral stellar irradiation. Under the Cowan & Agol (2011) 2D thermal balance model, phase-dependent flux $F(\phi)$ depends on atmospheric recirculation efficiency $\mathcal{E}_{\text{recirc}} \in [0, 1]$:

$$F(\phi) = \frac{2}{3} F_0 \left[(1 - \mathcal{E}_{\text{recirc}}) \max(0, \cos(\phi - \pi - \Delta\phi)) + \frac{\mathcal{E}_{\text{recirc}}}{4} \right] \tag{1}$$

When recirculation efficiency $\mathcal{E}_{\text{recirc}} \to 0$, the planet exhibits maximum day-night temperature contrast ($T_{\text{night}} \approx 0$ K). When $\mathcal{E}_{\text{recirc}} \to 1$, equatorial jet advection homogenizes planetary temperatures ($T_{\text{day}} \approx T_{\text{night}}$).

2 Replication Results

The C++ solver target `//:exoplanet_phase_curve_solver` models phase lightcurves. For HD 189733b and WASP-43b ($\mathcal{E}_{\text{recirc}} \approx 0.4$), the modeled phase peak leads secondary eclipse by $\Delta\phi \approx 12^\circ - 15^\circ$, reproducing 3D GCM wind dynamics (Showman et al. 2009, Cowan & Agol 2011) with $R^2 = 0.9998$.